RESEARCH hysteria2

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Deep web research for a Mullvad-parity self-hosted VPN spec. Topic: Hysteria2 + Salamander vs MASQUE; sing-box; AmneziaWG; udp2tcp/Shadowsocks; which obfuscation survives which censorship regime. All facts cite a source URL + access date. Web access: **YES** (WebSearch reachable; WebFetch blocked on a few domains — `v2.hysteria.network`, `mullvad.net` — worked around via mirrors/search summaries). Access date for all entries unless noted: **2026-06-25**.

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1. Hysteria2 + Salamander / Gecko obfuscation

What it is. Hysteria is a TCP & UDP proxy built on **QUIC (RFC 9000) with the Unreliable Datagram Extension (RFC 9221)**. The protocol shipped since v2.0.0 is internally "v4". It is designed for speed + censorship resistance, with a custom congestion-control called **Brutal** that deliberately *ignores packet loss* to maximise throughput on lossy/throttled links (the "bandwidth cheating" behaviour — the client tells the server a fixed bandwidth and Brutal sends at that rate regardless of loss signals). [hysteria protocol docs; sing-box hysteria2 docs]

Salamander obfuscation. Encapsulates *all* QUIC packets. The obfuscator computes a **BLAKE2b-256** hash of a randomly-generated **8-byte salt** appended to a **user-provided pre-shared key**, and XOR/scrambles every packet into seemingly-random bytes with no pattern. Purpose: defeat networks that specifically fingerprint/block **QUIC or HTTP/3** (but not UDP in general) — it hides the QUIC Initial packet structure so DPI cannot see a QUIC handshake or SNI. It does **not** help where UDP itself is blocked. [hysteria protocol docs]

Gecko (experimental). Builds on Salamander and *additionally fragments the QUIC handshake packets into randomly-sized, randomly-padded chunks* (configurable `min_packet_size` / `max_packet_size`). This directly targets the GFW's single-datagram QUIC-Initial inspection weakness (see §5). Added to both apernet hysteria and sing-box in 2025–2026. [sing-box changelog; hysteria changelog]

Current version (load-bearing fact): apernet/hysteria latest = **app/v2.9.2, released 2026-05-23**. That release adds the **Gecko** obfuscation, fixes a UDP ACL-bypass security issue, and OOM hardening. ("important security fixes … strongly encourage everyone to upgrade.") [github.com/apernet/hysteria/releases]

Auth. Hysteria2 uses a password/userpass over an HTTP/3-style auth exchange inside the QUIC tunnel; combined with Salamander PSK this gives two independent secrets (PSK for the wire obfuscation, password for authz).

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2. MASQUE (CONNECT-UDP, RFC 9298) — and Mullvad's 2025 deployment

What it is. RFC 9298 "Proxying UDP in HTTP" (Aug 2022) defines CONNECT-UDP: the client sends an Extended CONNECT with `:protocol = connect-udp`, encodes target host/port in a URI template, and the proxy maps **QUIC DATAGRAM frames ↔ UDP packets** to the target. Over HTTP/3 it is native-UDP; **RFC 9298 + RFC 9484 (CONNECT-IP) require HTTP/2 fallback** when QUIC is unavailable (works in maximally-restrictive nets, but loses native-UDP semantics → TCP-over-TCP-ish). [ietf-wg-masque draft / http.dev/masque]

Mullvad's production use (the parity target). Mullvad shipped **"QUIC Obfuscation for WireGuard"** on **2025-09-09, app version 2025.9**, on all desktop platforms (Android/iOS announced as forthcoming). It **tunnels WireGuard's UDP through an HTTP/3 proxy using MASQUE (RFC 9298)** so the traffic "resemble[s] standard web activity." Default behaviour: the desktop app **automatically tries QUIC after failed connection attempts** (can be forced permanently in settings/CLI). It is positioned for "regions/networks where WireGuard or Mullvad's other obfuscation methods face restrictions." Mullvad published **no performance/overhead numbers** and stated **no explicit threat it does not defend against** in the launch post. [mullvad blog 2025/9/9; alternativeto.net 2025-09]

Tooling. `quic-go/masque-go` is the reference Go implementation of RFC 9298 (MASQUE over HTTP/3) — directly relevant if building a self-hosted MASQUE proxy in Go. [github.com/quic-go/masque-go]

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3. sing-box protocol landscape (2025–2026)

sing-box supports the relevant censorship-resistant set: **Hysteria2, TUIC, VLESS+ Reality, ShadowTLS, AnyTLS, Shadowsocks, Trojan, WireGuard**. Key facts:

- **Hysteria2** — QUIC + Brutal; "measurably higher throughput than TCP-based alternatives on stable connections"; **Gecko** obfuscation added as a new QUIC obfs type with `min/max_packet_size`. Caveat: UDP-based, so **blocked in corporate/restricted nets that drop UDP** → best as one option in a fallback chain, not a sole solution. [sing-box hysteria2 docs; sing-box changelog]
- **VLESS + Reality** — the standout for DPI evasion: borrows the **TLS fingerprint of a real high-traffic site**, making connections indistinguishable from normal HTTPS to DPI. TCP-based → survives UDP blocks. [curevpn comparison; sing-box]
- **AnyTLS** (added to sing-box ~**March 2025**) — designed to mitigate TLS-proxy traffic-shape characteristics with a new multiplexing scheme. [sing-box changelog]
- **ShadowTLS** — wraps an inner protocol inside a real TLS handshake to a real site so the handshake passes as genuine TLS. [sing-box]

Architectural takeaway: a Mullvad-parity self-host wants **a QUIC/UDP primary (Hysteria2 or MASQUE) + a TLS-camouflage TCP fallback (Reality / ShadowTLS / AnyTLS) + a UDP-block fallback (udp2tcp/Phantun)** rather than one protocol.

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4. AmneziaWG — obfuscated WireGuard (the WireGuard-native alternative)

WireGuard fork adding protocol-level obfuscation against DPI. The standard WireGuard fingerprint DPI exploits: **fixed 32-bit message types (1–4) + invariant packet sizes** (handshake init always **148 bytes**, response always **92 bytes**). AmneziaWG breaks both:

- **Junk packets:** before each handshake, send `Jc` junk packets of random size `Jmin..Jmax` (noise; client-side recommended only).

- **Header/size obfuscation (AmneziaWG 2.0):** dynamic header ranges instead of static 1–4, random padding on all message types, extended signature/junk ("CPS") packets before handshakes to mimic other UDP protocols. **Every server effectively speaks its own dialect → no universal DPI signature.**
- **Performance:** ~3% throughput cost on an uncensored net (WireGuard 95 Mbps vs AmneziaWG 92 Mbps in a cited benchmark).
- Still **UDP-based** → does not survive a hard UDP block on its own (pair with udp2tcp/Phantun). [docs.amnezia.org/amnezia-wg; dev.to AmneziaWG 2.0; deepwiki amneziawg-go]

Relevance to a Mullvad-parity self-host: AmneziaWG is the *lowest-overhead* DPI evasion when the threat is **WireGuard fingerprinting, not UDP blocking**; MASQUE/ Hysteria2 are heavier but survive QUIC-permitting nets and HTTP-camouflage regimes.

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5. Which obfuscation survives which censorship regime

(a) China GFW — SNI-based QUIC censorship (USENIX Security 2025)

- GFW began **blocking QUIC by SNI since ~2024-04-07**; independently confirmed by USENIX Sec '25 ("Exposing and Circumventing SNI-based QUIC Censorship of the GFW") and UPB-SysSec (discovered **Jan 2025**, tested **Mar–Apr 2025**).
- Mechanism: GFW **derives the QUIC-Initial decryption key from the packet header** (the key is header-derivable per RFC 9001), extracts the **SNI**, matches a blocklist. Weekly it blocked an avg of **43.8K FQDNs** (Oct 2024–Jan 2025).
- **Critical weakness for circumvention:** the GFW **does NOT reassemble QUIC Initials split across >1 UDP datagram.** Chrome (since 2024-09-13) made Initials too large for one datagram → GFW less effective; Hysteria/sing-box/Xray added GFW-QUIC workarounds **almost immediately**. Firefox (Apr 2025) splits SNI to slip past. → **Gecko's handshake fragmentation directly defeats this.**
- Enforcement is **residual** (3-min block after a trigger; ~500ms to start; 58% 3-tuple / 37% 4-tuple). Long-lived QUIC tunnels are disrupted; short-lived may complete in the uncensored window. **Salamander (full QUIC scramble) hides the SNI/ Initial entirely → survives this regime**; plain QUIC does not.
- Researchers explicitly warn QUIC-reliant tools to "anticipate further GFW efforts to block QUIC and increased residual

censorship." [gfw.report/usenixsecurity25; upb-syssec.github.io/blog/2025/quic-china; theregister 2025-08-04]

(b) Hard UDP block / UDP throttling (common in China domestic + corporate)

- QUIC/Hysteria2/MASQUE-over-H3/WireGuard/AmneziaWG **all fail** — they need UDP.
- Survivors: **TCP-camouflage** (VLESS+Reality, ShadowTLS, AnyTLS) **or udp2tcp**: `dndx/phantun` masquerades UDP as a *fake* TCP stream through L3/L4 NAT/firewalls, **12-byte overhead**, no real TCP retransmit/flow-control penalty (user-mode TCP state machine, 100% safe Rust). Standard pairing: WireGuard/AmneziaWG **over Phantun** for UDP-blocked nets. MASQUE's **HTTP/2 fallback** is the in-protocol analogue. [github.com/dndx/phantun; deepwiki phantun; mullvad]

(c) SNI filtering / TLS-based DPI (TCP 443)

- Survivors: **VLESS+Reality** (real-site TLS fingerprint), **ShadowTLS** (real TLS handshake to a real host), **AnyTLS**. Salamander-obfuscated QUIC also has no SNI on the wire. Plain TLS proxies with a self-issued cert are detectable. [sing-box]

(d) Active probing (Shadowsocks-class)

- GFW passively flags suspicious connections (partly by **packet-size** distribution), then **actively probes the server** to confirm. Mitigation: randomise packet sizes; AEAD Shadowsocks + probe-resistant front. Hysteria2's password auth + Salamander make active probing return nothing useful (no protocol response to an unauthenticated prober). [gfw.report/blog/gfw_shadowsocks; gfw.report/blog/ss_advise]

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6. Hysteria2-primary vs MASQUE-primary — the tradeoff (current facts)

Axis	Hysteria2-primary (+ Salamander/Gecko)	MASQUE-primary (RFC 9298, à la Mullvad)
Standardisation	Custom protocol, single main impl (apernet) + sing-box	**IETF RFC 9298**, multiple impls (masque-go, Cloudflare, Apple Private Relay)

Axis	Hysteria2-primary (+ Salamander/ Gecko)	MASQUE-primary (RFC 9298, à la Mullvad)
Throughput on lossy/throttled links	**Best** — Brutal ignores loss, "bandwidth cheating"	Standard QUIC congestion control (fair, lower on lossy links)
Looks like normal web traffic	Salamander = random noise (hides QUIC, but *is* unusual UDP); Gecko fragments	**Strong** — genuinely *is* HTTP/3 to an HTTP proxy; blends with web
GFW QUIC-SNI (2024–25)	Survives via Salamander (no SNI) / Gecko (frag)	Survives **iff** the H3 proxy's own SNI is unblocked/fronted; otherwise SNI-filtered like any QUIC
Hard UDP block	Fails (needs UDP) — add Phantun	**Has in-protocol HTTP/2 fallback** (degraded but connects)
Self-host maturity	Very mature, one-click installers, low-RAM	Newer for self-host; masque-go exists; more moving parts
WireGuard compatibility	Tunnels arbitrary TCP/UDP incl. WG	**Designed to carry WireGuard UDP** (Mullvad's exact model)
Ecosystem clients	sing-box, mihomo, NekoBox, etc.	sing-box (less mature), Mullvad app, browsers' H3 stacks

Synthesis for a Mullvad-parity self-host:

- **Primary** should be a **QUIC/UDP obfuscated transport**. Mullvad chose **MASQUE** because it is an **IETF standard that genuinely looks like HTTP/3 web traffic** and carries WireGuard UDP — strongest "blend in" + in-protocol HTTP/2 fallback.
- **Hysteria2 + Salamander/Gecko** beats MASQUE on **raw throughput over throttled/lossy links** and on **out-of-the-box GFW-QUIC evasion (Gecko frag)**, with far more mature self-host tooling — but it is a non-standard protocol whose obfuscation is *random noise* rather than *genuine web traffic*.
- **Neither survives a hard UDP block alone** → both need a TCP fallback (MASQUE: native HTTP/2 fallback; Hysteria2: pair with **Phantun udp2tcp** or a Reality/ShadowTLS sibling).

- **Recommended layered design:** MASQUE **or** Hysteria2/Gecko as QUIC primary → AmneziaWG for low-overhead WG-fingerprint evasion where UDP is allowed → Phantun (udp2tcp) + VLESS-Reality/ShadowTLS as the UDP-blocked / SNI-filtered fallback chain. This mirrors Mullvad's "auto-try obfuscation after failure" model.

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